

SINGULAR KÄHLER–RICCI SHRINKERS AND POLARIZED FANO FIBRATIONS

ABSTRACT. We prove that every singular Kähler–Ricci shrinker arising as an $\mathbb{F}$ -limit of non-collapsed Kähler–Ricci flows admits a polarized Fano fibration. In particular, its complex analytic space is the analytification of a normal quasi-projective variety.

1. INTRODUCTION

It is proved in [SZ26, Theorem 1.1] that every smooth Kähler–Ricci shrinker naturally defines a polarized Fano fibration. The notion of a polarized Fano fibration was introduced in [SZ26] and is recalled as Definition 2.1 in this paper. We prove the corresponding statement for singular shrinkers arising as $\mathbb{F}$ -limits of non-collapsed Kähler–Ricci flows.

Theorem 1.1. *Let X be a singular Kähler–Ricci shrinker arising as an $\mathbb{F}$ -limit of non-collapsed Kähler–Ricci flows. Then X admits a polarized Fano fibration. In particular, its underlying complex analytic space is the analytification of a normal quasi-projective variety.*

The analytic foundation for the proof of Theorem 1.1 is provided by Hallgren–Zhang [HZ26], who proved that X is a normal analytic space with log terminal singularities, constructed the complexified torus action and its proper moment map, and established the projectivity and klt property of the reductions at regular values. As in [SZ26], the crucial point is to show that the vanishing locus of the soliton vector field, or equivalently, the fixed-point locus of the compact torus $\mathbb{T}$ generated by ξ , is compact. To this end, we perform Kähler reduction at regular values and study the birational transformations that occur when crossing critical values, a phenomenon commonly referred to as wall crossing.

In the smooth setting considered in [SZ26], the coefficients of the difference divisors arising from wall crossing can be computed explicitly using local models of the quotients. The new difficulty here is that the fixed center of the wall is singular, in which case it appears difficult to determine these coefficients explicitly. The key observation is that their precise values are not needed: as long as the difference divisors have the correct sign, both the propagation of complements and the preservation of a lower bound for discrepancies across the wall can still be established without an explicit local normal form.

Let $c > 0$ be a critical value, and choose $\varepsilon > 0$ sufficiently small so that

$$X_- := X_{c-\varepsilon}, \quad X_+ := X_{c+\varepsilon}$$

are nonempty adjacent regular reductions. Let X_c be the critical quotient. Hallgren–Zhang supply the proper adjacent modifications $\pi_\pm: X_\pm \rightarrow X_c$ [HZ26, Proposition 6.9]. Let W be the normalization of the graph of the wall-crossing map, with projections $p_\pm: W \rightarrow X_\pm$. Thus

$$\begin{array}{ccccc} & & W & & \\ & \swarrow p_- & \downarrow g & \searrow p_+ & \\ X_- & \xrightarrow{\pi_-} & X_c & \xleftarrow{\pi_+} & X_+ \end{array} \tag{1.1}$$

and

$$g = \pi_- \circ p_- = \pi_+ \circ p_+.$$

Let $D_{\pm}$ be the quotient boundaries on $X_{\pm}$, and let $L_{\pm}$ be the corresponding rational line bundles obtained by descent from the character linearization; cf. [Tha96, §1, proof of Proposition 1.6, p. 695]. Although the classes $K_{X_{\pm}} + D_{\pm}$ and $L_{\pm}$ need not descend separately to the critical quotient, their stabilizer characters cancel in the combination $K_{X_{\pm}} + D_{\pm} - cL_{\pm}$. After clearing denominators, this combination does descend, and pulling back gives the identity of $\mathbb{Q}$ -Cartier divisors bundles

$$p_-^*(K_{X_-} + D_- - cL_-) = p_+^*(K_{X_+} + D_+ - cL_+).$$

It remains to determine the sign of the character jump

$$E_c = p_-^*L_- - p_+^*L_+.$$

The two graded half-algebras of the wall show that $g_*E_c \geq 0$, while relative ampleness makes $-E_c$ nef over the critical quotient. The negativity lemma then gives $E_c \geq 0$ on the whole graph.

We can then proceed as in [SZ26] to establish the existence of complements and a uniform lower bound for discrepancies. Another new observation, compared with [SZ26], is that the coefficients of the boundary divisors belong to a finite set, which allows us to apply the boundedness result of Hacon–Xu [HX15]. We then obtain a uniform lower bound for the normalized volumes. Combined with finite-quotient estimates, this yields uniform bounds on both the stabilizers and the Cartier indices of the character line bundles in a single step. Finally, we use the relative length estimate of [BCHM10] to obtain a uniform upper bound for all critical values.

2. SETUP AND ANALYTIC INPUTS

Let (X, d) be a singular Kähler–Ricci shrinker arising as an $\mathbb{F}$ -limit of non-collapsed Kähler–Ricci flows. By [HZ26, Theorem A], X is a normal complex analytic variety with log terminal singularities, and its metric regular locus agrees with its complex analytic smooth locus. We denote this common open set by X^{reg} and write (ω, J, f) for the smooth (incomplete) Kähler–Ricci shrinker structure on X^{reg} . Moreover, ω extends to X as a positive $(1, 1)$ -current with bounded local potentials. The soliton field

$$\xi := J\nabla f$$

generates a compact torus $\mathbb{T}$ acting by biholomorphic isometries, and $\mathbb{T}^{\mathbb{C}}$ acts holomorphically on X by [HZ26, Section 5 and Proposition 5.3].

For a normal analytic or algebraic space V and $m \in \mathbb{Z}$, we write $\mathcal{O}_V(mK_V)$ for the m -th reflexive canonical power, using the reflexive dual when $m < 0$. When a group acts on V , this sheaf carries the canonical pullback linearization induced on V^{reg} and extended reflexively.

Definition 2.1 (Polarized Fano fibration). A polarized Fano fibration is a triple $(\pi: X \rightarrow Y, \xi)$ with the following properties. The spaces X and Y are normal quasi-projective varieties, π is surjective and projective, $\pi_*\mathcal{O}_X = \mathcal{O}_Y$, the variety X is klt, and $-K_X$ is $\mathbb{Q}$ -Cartier and relatively ample. There is a compact torus $\mathbb{T}$ acting on X and Y such that π is T -equivariant, $Y = \text{Spec } R$ is affine with a unique $\mathbb{T}$ -fixed point, and $\zeta \in \text{Lie}(\mathbb{T})$ lies in the Reeb cone: if

$$R = \bigoplus_{\alpha} R_{\alpha}$$

is the $\mathbb{T}$ -weight decomposition, then $\langle \alpha, \zeta \rangle > 0$ for every nonzero weight with $R_{\alpha} \neq 0$.

2.1. Analytic inputs. Hallgren–Zhang supply the analytic reduction package used throughout. X is a normal analytic space with log terminal singularities and [HZ26, Theorem A(ii) and Proposition 4.6] shows that ω has continuous strictly plurisubharmonic local potentials and is bounded below, in local embeddings, by the ambient Euclidean Kähler form. Moreover the canonical moment map is continuous and proper, with closed convex image [HZ26, Theorem 6.3]; its levels and compact slabs are connected [HZ26, Proposition 6.5].

We may henceforth assume that X is noncompact and $n := \dim_{\mathbb{C}} X \geq 2$. Indeed, suppose that X is compact. By [HZ26, Theorem A(ii) and Proposition 4.6], ω has continuous strictly plurisubharmonic local potentials and is bounded below, in local embeddings, by the ambient Euclidean Kähler form. Choose $m > 0$ such that $-mK_X$ is Cartier, and set $L := \mathcal{O}_X(-mK_X)$. The current $m\omega$ defines a positive continuous Hermitian metric on L ; Richberg regularization and Grauert's extension of the Kodaira embedding theorem [Gra62, §3, Satz 2 and its proof, pp. 343–345] show that L is ample. Thus X is projective Fano and $(X \rightarrow \{\text{pt}\}, \xi)$ is a polarized Fano fibration.

By [HZ26, Theorem 6.3], the $\mathbb{T}$ -action has a continuous proper moment map

$$\mu: X \longrightarrow \text{Lie}(\mathbb{T})^*$$

with closed convex image. Its additive normalization is fixed by the canonical lift of the action to $\mathcal{O}_{X^{\text{reg}}}(-K_{X^{\text{reg}}})$. Namely, on X^{reg} ,

$$u_\eta = \frac{1}{2}(\text{div}(J\eta) - J\eta(f)), \quad \iota_\eta \omega = -du_\eta. \quad (2.1)$$

Since $J\xi = -\nabla f$, equations (2.6)–(2.7) of [HZ26] give

$$\begin{aligned} u_\xi &= -\Delta f + |\nabla^{1,0} f|^2 \\ &= R - n + |\nabla^{1,0} f|^2 = f - (n + W). \end{aligned}$$

Thus the soliton component of the canonical moment map is the proper function f shifted by the fixed number $n + W$, where W is the shrinker entropy constant in equation (2.7) of [HZ26].

By the convexity argument of [SZ26, Lemma 3.1], there is a rational element $\eta_0 \in \text{Lie}(\mathbb{T})$, arbitrarily close to ξ , such that u_{η_0} is proper and bounded below. Indeed we have the following result.

Lemma 2.2. *There is a constant $C = C(\eta_0)$ such that*

$$C^{-1}f - C \leq u_{\eta_0} \leq C(1 + f - \min_X f).$$

After a positive rescaling, we may assume that

$$\eta := a\eta_0$$

is the primitive integral generator of the corresponding effective circle. We write

$$u := u_\eta = \langle \mu, \eta \rangle.$$

With this choice all characters are integral and no period factor enters the weight formulas.

Let ρ_λ be the induced point action of $\mathbb{C}^*$, normalized so that $\rho_{e^{\sqrt{-1}\theta}}$ is generated by η . Holomorphicity of the action and $\eta = J\nabla u$ give

$$\left. \frac{d}{ds} \right|_{s=0} \rho_{e^s} = -J\eta = \nabla u, \quad \frac{d}{ds} u(\rho_{e^s} x) = |\nabla u|^2(\rho_{e^s} x). \quad (2.2)$$

Thus $\lambda \rightarrow 0$ is the decreasing- u direction and $\lambda \rightarrow \infty$ is the increasing- u direction. Since X is connected, properness and noncompactness give

$$u(X) = [\min_X u, \infty),$$

so every critical value above the minimum has nonempty levels on both sides.

On real-valued functions on X^{reg} set

$$\Delta_f := \Delta - \frac{1}{2} \langle \nabla f, \nabla \cdot \rangle.$$

This is the complex drift Laplacian, symmetric for $d\nu := e^{-f}\omega^n$. Differentiating $\iota_\eta \omega = -du_\eta$ and using the shrinker equation gives

$$\Delta u_\eta + u_\eta + \frac{1}{2} J\eta(f) = 0.$$

Since $\nabla u_\eta = -J\eta$, the canonical potential satisfies

$$\Delta_f u_\eta + u_\eta = 0. \quad (2.3)$$

Lemma 2.3. *The canonical Hamiltonian satisfies*

$$u_\eta, \nabla u_\eta \in L^2(X, d\nu), \quad \int_X u_\eta e^{-f} \omega^n = 0.$$

Proof. The potential estimate and volume-ratio upper bound for a non-collapsed $\mathbb{F}$ -limit shrinker [CMZ24, Theorem 1.1 and Corollaries 1.2–1.3] imply

$$\int_X (1 + f - \min_X f)^k e^{-f} \omega^n < \infty \quad \text{for every fixed } k.$$

Lemma 2.2 therefore gives $u_\eta \in L^2(d\nu)$.

To integrate by parts across the singular set and at infinity, let χ_ϵ be the singular-set cutoffs of [HZ26, Lemma 3.3], and let ρ_R be the proper- f cutoffs of [HZ26, Lemma 3.4], so that $\rho_R = 1$ on $\{f \leq R^2\}$, it is supported in $\{f \leq 2R^2\}$, and

$$|\nabla \rho_R| \leq CR^{-1}, \quad |\Delta \rho_R| \leq CR^{-2}.$$

Put $\psi_{\epsilon,R} := \chi_\epsilon \rho_R$. Multiplying (2.3) by $\psi_{\epsilon,R}^2 u_\eta$ and integrating by parts gives the Caccioppoli estimate

$$\int_{X^{\text{reg}}} \psi_{\epsilon,R}^2 |\nabla u_\eta|^2 d\nu \leq C \int_{X^{\text{reg}}} (\psi_{\epsilon,R}^2 + |\nabla \psi_{\epsilon,R}|^2) u_\eta^2 d\nu.$$

For fixed R , choose $\sigma \in (0, 2)$. The χ_ϵ error tends to zero: its gradient tends to zero in $L^{4-\sigma}$ on compact sets and hence in L^2 there, while u_η is locally bounded. After letting $\epsilon \downarrow 0$, the ρ_R error tends to zero as $R \rightarrow \infty$ by its derivative bound, the growth estimate for u_η , and the finite weighted moments of f . The Caccioppoli estimate consequently gives $\nabla u_\eta \in L^2(d\nu)$.

We now integrate (2.3) against $\psi_{\epsilon,R}$. Weighted integration by parts yields

$$\int_{X^{\text{reg}}} \psi_{\epsilon,R} u_\eta d\nu = \frac{1}{2} \int_{X^{\text{reg}}} \langle \nabla \psi_{\epsilon,R}, \nabla u_\eta \rangle d\nu,$$

where the factor records the complex-Laplacian normalization. The right-hand side tends to zero, first as $\epsilon \downarrow 0$ and then as $R \rightarrow \infty$, by the cutoff estimates and the L^2 bound just proved. The left-hand side converges to $\int_X u_\eta d\nu$, proving the asserted centering. $\square$

2.2. Reductions at regular values. For a smooth total space this is the usual orbifold reduction used in [SZ26, Section 3]. In the singular setting, Hallgren–Zhang’s algebraic quotient theorem replaces the smooth quotient construction. Fix a regular value r . By [HZ26, Theorem 6.6], every regular reduction is a normal projective variety of klt type and invariant neighborhoods admit equivariant quasi-projective algebraizations [HZ26, Theorem B(iii)] such that the analytic Hilbert quotient of $X^s(\eta, r)$ agrees with its algebraic good quotient. We write

$$X_r := X^s(\eta, r)/\mathbb{C}^*.$$

Its quotient boundary is

$$D_r := \sum_B \left(1 - \frac{1}{e_B}\right) B,$$

where e_B is the stabilizer order of the general orbit above B ; terms with $e_B = 1$ are omitted. For different r , X_r are birational to each other and we identify their function fields.

Put $U_r := X^s(\eta, r)$. We use the geometric pullback convention: a function or form has weight k if

$$\rho_t^* f = f \circ \rho_t = t^k f \quad (t \in \mathbb{C}^*).$$

Let $\tilde{L}_r := \mathcal{O}_{U_r} e$ be the trivial line bundle with chosen nowhere-vanishing global frame e , and equip it with the inverse-standard $\mathbb{C}^*$ -linearization. Explicitly, for each $t \in \mathbb{C}^*$, let

$$\lambda_t: \rho_t^* \tilde{L}_r \xrightarrow{\sim} \tilde{L}_r$$

be the bundle isomorphism over U_r determined by

$$\lambda_t(\rho_t^* e) = t^{-1} e.$$

Thus ρ_t is the point action on U_r , whereas λ_t is its linearization on the trivial line bundle, with fiber character t^{-1} . Consequently, if f has pullback weight k , then $f e^{\otimes k}$ is invariant.

Choose m divisible by all stabilizer orders. Such an m exists because the compact quotient is covered by finitely many finite-stabilizer slices. The descent theorem for equivariant line bundles [KKV89, §4, Proposition 4.2, pp. 83–84] then gives a line bundle $L_r^{(m)}$ on X_r whose pullback is the linearized line bundle $\tilde{L}_r^{\otimes m}$. Set

$$L_r := \frac{1}{m} L_r^{(m)} \in \text{Pic}(X_r)_{\mathbb{Q}}.$$

If m and m' are two such choices, descent of the common multiple $\text{lcm}(m, m')$, together with uniqueness of descent, shows that $m^{-1} L_r^{(m)} = (m')^{-1} L_r^{(m')}$; hence L_r is independent of m . The following result follows from [HZ26, Theorem 6.6 and its proof and Proposition 6.8]

Proposition 2.4. *For every regular value r , (X_r, D_r) is klt, and there exists a Kähler current ω_r with local continuous potentials inside the following Kähler class*

$$A_r := -(K_{X_r} + D_r) + r L_r \in \text{Amp}(X_r). \quad (2.4)$$

2.3. The zero-level patch and the first complement.

Lemma 2.5. *There is a point*

$$x \in X^{\text{reg}} \setminus \text{Fix}(S^1) \quad \text{such that} \quad u(x) = 0, \quad du(x) \neq 0.$$

Its S^1 -isotropy group is finite.

Proof. Lemma 2.3 then implies that u takes both positive and negative values. Since $X^{\text{reg}} \setminus \text{Fix}(S^1)$ is dense, both sign sets meet this locus. It is also path-connected: it is the set $X_{\max}(\eta)$ appearing in the proof of [HZ26, Proposition 6.5]. A path joining the two sign points therefore meets $\{u = 0\}$ at a point x with $\eta(x) \neq 0$. Hence

$$du(x) = -\iota_{\eta(x)} \omega \neq 0.$$

The stabilizer of x is a proper closed subgroup of S^1 , and is therefore finite. $\square$

Lemma 2.6. *There are $\varepsilon > 0$, compact quotient patches $K_r \subset X_r$ varying smoothly for regular $|r| < \varepsilon$, and a number $v > 0$ such that*

$$\int_{K_r} \omega_r^{n-1} \geq v.$$

Consequently,

$$A_r^{n-1} \geq v.$$

Proof. Apply the locally free Hamiltonian normal form at the point x of Lemma 2.5. After shrinking an invariant neighborhood of the compact orbit $S^1 x$, every nearby level is locally free, and its orbifold quotient is identified with a fixed transverse slice. Choose a relatively compact domain with smooth boundary in this slice and let K_r be its closed image in $u^{-1}(r)/S^1 \subset X_r$. The reduced forms ω_r depend smoothly on r on this family of compact orbifold domains. Since K_0 has nonempty interior,

$$\int_{K_0} \omega_0^{n-1} > 0.$$

Continuity gives the asserted lower bound after decreasing ε .

The current ω_r represents $2\pi c_1(A_r)$ and has continuous local potentials. By pulling back to a resolution, we know that its Bedford–Taylor product is defined and

$$A_r^{n-1} = (2\pi)^{-(n-1)} \int_{X_r} (\omega_r^{n-1})_{\text{BT}}.$$

It follows that

$$A_r^{n-1} \geq (2\pi)^{-(n-1)} \int_{K_r} \omega_r^{n-1}.$$

After decreasing v by the fixed factor $(2\pi)^{-(n-1)}$, the result follows. $\square$

The critical values are discrete by [HZ26, Lemma 6.4]. They are also closed, since u is proper on the closed fixed locus. After decreasing ε , there is no critical value in $(0, \varepsilon)$. The reductions in this chamber are naturally identified.

Proposition 2.7. *On the first positive reduction there are an effective big $\mathbb{Q}$ -divisor Δ_r and a fixed number $\delta > 0$ such that*

$$K_{X_r} + D_r + \Delta_r \sim_{\mathbb{Q}} 0, \quad (X_r, D_r + \Delta_r) \text{ is } \delta\text{-lc}.$$

Proof. We write

$$(X_r, D_r, L_r) = (X_+, D, L), \quad H := -(K_{X_+} + D).$$

On the fixed projective variety X_+ , Proposition 2.4 implies

$$A_r = -(K_{X_r} + D_r) + rL_r = H + rL$$

is ample. Letting $r \downarrow 0$ proves that H is nef. By Lemma 2.6 and continuity of top intersection,

$$H^{n-1} = \lim_{r \downarrow 0} A_r^{n-1} \geq v > 0.$$

Therefore the nef divisor H is big.

The pair (X_+, D) is klt. Choose a sufficiently divisible integer $m \geq 2$ such that mH is Cartier. By the klt base-point-free theorem, after replacing m by a sufficiently divisible multiple, $|mH|$ is base-point-free. Let $G \in |mH|$ be general and set

$$\Delta := \frac{1}{m}G.$$

Then Δ is effective and big, and

$$K_{X_+} + D + \Delta \sim_{\mathbb{Q}} 0.$$

For a general G , Bertini theorem implies that $(X_+, D + \Delta)$ is klt. Choose $N > 0$ such that $N(K_{X_+} + D + \Delta)$ is Cartier. Its log discrepancies belong to $N^{-1}\mathbb{Z}_{>0}$, so the pair is N^{-1} -lc. Set $\delta := N^{-1}$. $\square$

3. WALL CROSSING ON A SINGULAR TOTAL SPACE

Let $c > 0$ be a critical value of u with both adjacent levels nonempty, and choose $\varepsilon > 0$ so that $c \pm \varepsilon$ are regular values with nonempty levels and c is the unique critical value in $(c - \varepsilon, c + \varepsilon)$. Write

$$U_- := X^s(\eta, c - \varepsilon), \quad U_+ := X^s(\eta, c + \varepsilon), \quad U_c := X^{ss}(\eta, c), \quad \varpi_c: U_c \longrightarrow X_c,$$

$$X_- := X_{c-\varepsilon}, \quad X_+ := X_{c+\varepsilon}.$$

By [HZ26, Proposition 6.9], there are proper modifications $\pi_{\pm}: X_{\pm} \rightarrow X_c$ to a compact quotient X_c ; these are the adjacent maps in diagram (1.1). There is one change of convention. The radial parameter used in the proof of that proposition has generator $J\eta$, whereas our parameter λ has generator $-J\eta$ by (2.2). Thus Z_i^- in [HZ26, Proposition 6.9] is the $\lambda \rightarrow \infty$ stratum on X_- , while Z_i^+ is the $\lambda \rightarrow 0$ stratum on X_+ .

The wall-crossing argument has two ingredients, both of which avoid the need for a smooth transverse model at the limiting fixed component. The first is critical descent. Using the canonical-weight formula proved below, we construct a line bundle $A_{c,N}$ on X_c such that

$$\mathcal{O}_{X_\pm}(N(K_{X_\pm} + D_\pm - cL_\pm)) \simeq \pi_\pm^* A_{c,N}.$$

The second ingredient is the effectivity of the character jump

$$E_c := p_-^* L_- - p_+^* L_+,$$

proved by combining $g_* E_c \geq 0$ from the graded wall algebra, the g -nefness of $-E_c$ from relative ampleness, and the negativity lemma.

3.1. Canonical weights at fixed components. We first record the relation between the canonical character at a fixed component and the normalized Hamiltonian. Let Z be an irreducible component of the fixed locus of the $\mathbb{C}^*$ -action generated by η . By [HZ26, Lemma 6.4], the Hamiltonian is constant on Z ; denote its value by c_Z . Choose a general point $z \in Z$ and an integer $m > 0$ such that mK_X is Cartier near z . The canonical pullback linearization makes the one-dimensional fiber $(\mathcal{O}_X(mK_X))_z$ a $\mathbb{C}^*$ -representation. Denote its weight by $a_{Z,m}$, so for every $0 \neq v \in (\mathcal{O}_X(mK_X))_z$,

$$\lambda_t^{K,m}(\rho_t^* v) = t^{a_{Z,m}} v.$$

Lemma 3.1 (Canonical weight formula). *One has $c_Z \in \mathbb{Q}$ and*

$$a_{Z,m} = -mc_Z.$$

Proof. The proof of [HZ26, Lemma 6.4, equation (6.7)] applies at an arbitrary fixed point, including a singular one. It gives $\ell > 0$ such that $-\ell K_X$ is Cartier near z , together with a nonvanishing local eigengenerator s of $\mathcal{O}_X(-\ell K_X)$ with character β_Z , such that

$$c_Z = u(z) = \frac{\beta_Z}{\ell}.$$

Thus $c_Z \in \mathbb{Q}$. Fix $0 \neq v \in (\mathcal{O}_X(mK_X))_z$, and let M be a common multiple of m and ℓ . The nonzero vectors

$$s(z)^{M/\ell}, \quad (v^\vee)^{M/m}$$

lie in the same one-dimensional fiber $(\mathcal{O}_X(-MK_X))_z$, so their characters agree. Hence

$$\frac{M}{\ell} \beta_Z = -\frac{M}{m} a_{Z,m},$$

and therefore $a_{Z,m} = -mc_Z$. □

3.2. Critical descent and the graph identity. We now apply the weight formula to the critical value c fixed at the beginning of the section. Lemma 3.1 gives $c \in \mathbb{Q}$. Let

$$U_c := X^{ss}(\eta, c), \quad \varpi_c: U_c \longrightarrow X_c$$

be the critical semistable locus and its analytic Hilbert quotient, and let $\tilde{L}_c := \mathcal{O}_{U_c} e$ carry the inverse-standard linearization fixed above. For any positive integer N with $Nc \in \mathbb{Z}$ and NK_{U_c} Cartier, put

$$\mathcal{N}_{c,N} := \mathcal{O}_{U_c}(NK_{U_c}) \otimes \tilde{L}_c^{\otimes(-Nc)}.$$

Applying Lemma 3.1 with $m = N$, the canonical factor has weight $-Nc$, which cancels the weight $+Nc$ of the inverse-standard character factor. The proof below chooses N further divisible so that every finite nonfixed stabilizer also acts trivially on $\mathcal{N}_{c,N}$.

Lemma 3.2 (Critical descent). *The integer N can be chosen so that the descent theorem gives a unique line bundle $A_{c,N}$ on X_c and an equivariant isomorphism*

$$\mathcal{N}_{c,N} \simeq \varpi_c^* A_{c,N}.$$

For this choice one has

$$\mathcal{O}_{X_\pm}(N(K_{X_\pm} + D_\pm - cL_\pm)) \simeq \pi_\pm^* A_{c,N}. \quad (3.1)$$

Proof. Since c is critical, $u^{-1}(c)$ contains a fixed point; hence Lemma 3.1 gives $c \in \mathbb{Q}$. The results of [HZ26, Theorem 6.6, Proposition 6.9, and the proofs of Proposition 6.11 and Corollary 6.12], together with [Gre10, Theorem 3], allow us to work algebraically: U_c is quasi-projective, ϖ_c is a good quotient onto the projective variety X_c , and the algebraic maps $\pi_\pm$ satisfy $\varpi_c|_{U_\pm} = \pi_\pm \circ \varpi_\pm$. Since U_c is klt and quasi-projective, a finite cover by Cartier-index neighborhoods gives $q > 0$ such that qK_{U_c} is Cartier.

Put

$$V := U_c \setminus U_c^{\mathbb{C}^*}, \quad I_V := \{(t, x) \in \mathbb{C}^* \times V : t \cdot x = x\}.$$

I_V is closed in $\mathbb{C}^* \times V$, hence is of finite type and quasi-compact. Its projection to V is quasi-finite, with geometric fibers the finite stabilizers μ_e . Their orders are therefore bounded by some B [SP, Tag 03JA]. Write $c = a/d$ in lowest terms and set

$$M := \text{lcm}\{1, \dots, B\}, \quad N := qdM.$$

Then NK_{U_c} is Cartier, $Nc \in \mathbb{Z}$, and every such e divides both $N/q = dM$ and $Nc = qaM$.

We verify the closed-orbit stabilizer criterion for $\mathcal{N}_{c,N}$. If $x \in U_c$ is fixed, then $u(x) = c$. Lemma 3.1 and local constancy of the character on a fixed component give weights $-Nc$ and $+Nc$ on the fibers of $\mathcal{O}_{U_c}(NK_{U_c})$ and $\tilde{L}_c^{\otimes(-Nc)}$, respectively. If a closed orbit is nonfixed, its stabilizer is $H = \mu_e$ and

$$\mathcal{O}_{U_c}(NK_{U_c}) \simeq \mathcal{O}_{U_c}(qK_{U_c})^{\otimes N/q}.$$

The divisibilities $e \mid N/q$ and $e \mid Nc$ trivialize the actions on the two factors. Thus every closed-orbit stabilizer acts trivially on $\mathcal{N}_{c,N}$, and the descent theorem [KKV89, §4, Proposition 4.2, pp. 83–84] gives a unique line bundle $A_{c,N}$ on X_c with

$$\mathcal{N}_{c,N} \simeq \varpi_c^* A_{c,N}.$$

It remains to identify the restricted descent. At a closed orbit in $U_\pm$ with finite stabilizer H , Luna’s algebraic étale slice theorem [Lun73] gives a normal slice S and strongly étale maps

$$\mathbb{C}^* \times_H S \longrightarrow U_\pm, \quad S/H \longrightarrow X_\pm.$$

The H -action on S is faithful: a kernel element fixes the associated bundle, since $[ht, z] = [t, hz] = [t, z]$, and hence a nonempty open subset of the effective irreducible $\mathbb{C}^*$ -space X . The invariant form dt/t removes the canonical factor in the orbit direction, so it remains to consider the finite quotient $\tau: S \rightarrow S/H$.

For a prime divisor $E \subset S$ above $B \subset S/H$, let e be the order of its inertia group. After localizing at the generic points of E and B , strong étaleness identifies e with the generic stabilizer order e_B defining the quotient boundary, and finite Riemann–Hurwitz gives

$$\tau^* B = eE, \quad R_\tau = (e - 1)E, \quad K_S = \tau^*(K_{S/H} + (1 - e^{-1})B).$$

This includes the pseudoreflexion case. Summing over the branch divisors yields $K_S = \tau^*(K_{S/H} + D_{S/H})$, where $D_{S/H} := \sum_B (1 - e_B^{-1})B$ is the pullback of $D_\pm$. Since N/q is divisible by every closed-orbit stabilizer order on $U_\pm$, the canonical factor descends; the formula above and reflexive extension identify its descent with

$$\mathcal{O}_{X_\pm}(N(K_{X_\pm} + D_\pm)).$$

Since Nc has the same divisibility, the definition of $L_\pm$ identifies the other descent as $\mathcal{O}_{X_\pm}(-NcL_\pm)$. Hence the descent of $\mathcal{N}_{c,N}|_{U_\pm}$ is

$$\mathcal{O}_{X_\pm}(N(K_{X_\pm} + D_\pm - cL_\pm)).$$

On the other hand, restricting the descent from X_c gives $\mathcal{N}_{c,N}|_{U_\pm} \simeq \varpi_\pm^* \pi_\pm^* A_{c,N}$. Uniqueness of descent proves (3.1). $\square$

Let W and $p_\pm$ be as in diagram (1.1). On the common stable quotient the canonical divisors and character lines are identified, but this does not determine their orders along divisors of W exceptional over X_c . Critical descent gives the stronger comparison.

Proposition 3.3 (Graph wall identity). *One has*

$$p_-^*(K_{X_-} + D_- - cL_-) = p_+^*(K_{X_+} + D_+ - cL_+) \quad (3.2)$$

as $\mathbb{Q}$ -Cartier divisors on W .

Proof. Let Ω be the common dense quotient open in X_- and X_+ . Choose one nonzero rational top form κ on Ω and one nonzero rational frame ℓ of the common character line there. Extend them rationally to $\kappa_\pm$ and $\ell_\pm$ on the two adjacent quotients. Since N and Nc clear the stabilizer denominators, the canonical rational sections of $\mathcal{O}_{X_\pm}(ND_\pm)$, together with $\kappa_\pm^{\otimes N}$ and $\ell_\pm^{\otimes (-Nc)}$, form rational sections $\sigma_\pm$ of $\mathcal{O}_{X_\pm}(N(K_{X_\pm} + D_\pm - cL_\pm))$. The quotient boundaries, top forms, and character frames agree on Ω , so $\sigma_-|_\Omega = \sigma_+|_\Omega$.

Transport $\sigma_\pm$ through the two isomorphisms of Lemma 3.2. Since $\pi_\pm$ are birational, there are unique rational sections $s_\pm$ of $A_{c,N}$ whose pullbacks are the transported sections. The equality of σ_- and σ_+ on the common dense quotient implies that $s_- = s_+$ in the generic fiber of $A_{c,N}$. Hence $s_- = s_+$. Denote this one rational section by s . Pulling s to W gives

$$\operatorname{div}_W((\pi_- \circ p_-)^* s) = Np_-^*(K_{X_-} + D_- - cL_-) = Np_+^*(K_{X_+} + D_+ - cL_+).$$

Dividing by N proves (3.2). $\square$

3.3. The graded character jump and effectivity. It remains to control the sign of the character jump. Let Ω be the common dense quotient open in X_- and X_+ , and identify the restrictions of L_- and L_+ there through the common equivariant inverse-character line on the stable locus. Choose an integer $m > 0$ for which mL_- and mL_+ are Cartier, and fix one nonzero rational frame $\ell^{(m)}$ of their common restriction to Ω . Denote its rational extensions to X_- and X_+ by $\ell_-^{(m)}$ and $\ell_+^{(m)}$, respectively. For every prime divisor $F \subset W$, define

$$\operatorname{coeff}_F(E_c) := \frac{1}{m} \left(\operatorname{ord}_F(p_-^* \ell_-^{(m)}) - \operatorname{ord}_F(p_+^* \ell_+^{(m)}) \right). \quad (3.3)$$

Equivalently, mE_c is the divisor of the rational section

$$p_-^* \ell_-^{(m)} \otimes (p_+^* \ell_+^{(m)})^{-1}$$

of $p_-^*(mL_-) \otimes p_+^*(mL_+)^{-1}$. Replacing $\ell^{(m)}$ by $h\ell^{(m)}$, for $h \in K(X_c)^\times = K(\Omega)^\times$, adds $\operatorname{ord}_F(g^*h)$ to both terms in (3.3); passing to a tensor power has the same cancellation. Thus the divisor is independent of these choices and is the rationally identified difference

$$E_c = p_-^* L_- - p_+^* L_+. \quad (3.4)$$

Lemma 3.4 (Affine description of the two adjacent quotients). *Let $V = \operatorname{Spec} R \subset X_c$ be affine and let*

$$\tilde{V} := \varpi_c^{-1}(V) = \operatorname{Spec} A, \quad A = \bigoplus_{j \in \mathbb{Z}} A_j, \quad A_0 = R,$$

where the grading is the geometric-pullback grading $\rho_\lambda^* f = \lambda^j f$ on A_j . Then

$$U_- \cap \tilde{V} = \bigcup_{\substack{j < 0 \\ f \in A_j}} D(f), \quad U_+ \cap \tilde{V} = \bigcup_{\substack{j > 0 \\ f \in A_j}} D(f).$$

For every sufficiently divisible $N > 0$, if

$$S_\pm := \bigoplus_{k \geq 0} A_{\pm kN},$$

then the two base-changed chamber quotients are

$$\begin{aligned} X_- \times_{X_c} V &= \text{Proj}_R S_-, \\ X_+ \times_{X_c} V &= \text{Proj}_R S_+. \end{aligned} \tag{3.5}$$

After enlarging N to clear the finite stabilizer characters, the tautological bundles satisfy

$$\mathcal{O}_{\text{Proj } S_-}(1) = -NL_-, \quad \mathcal{O}_{\text{Proj } S_+}(1) = NL_+.$$

Proof. The real subgroup $\lambda > 0$ raises u by (2.2); hence the lower chamber uses the negative character and the upper chamber uses the positive character. The affine Hilbert–Mumford criterion [Tha96, §1, Lemma 1.2 and Proposition 1.3] then gives the displayed unions: the limit at zero exists exactly when all negative-weight functions vanish, since negative weights obstruct the zero limit, and the analogous statement at infinity uses the positive weights, since positive weights obstruct the infinity limit. Thus lower stability means that the zero-limit does not exist, equivalently some negative-weight function is nonzero; upper stability is the analogous assertion at infinity for a positive-weight function.

By [Tha96, §1, p. 694], the character quotients of these opens are the two graded relative Proj schemes in (3.5). Sufficiently divisible common Veronese subrings are generated in degree one by [SP, Tag 0EGH], and replacing a ring by its Veronese preserves Proj and tracks its tautological twists by [SP, Tag 0B5J]. The final paragraph of the proof of [HZ26, Theorem 6.6], applying [Gre10, Proposition 10.1], makes each regular chamber quotient an algebraic Hilbert quotient. By [HZ26, Corollary 6.12], the maps $\pi_\pm$ are algebraic; restriction and uniqueness of good quotients give the asserted base changes.

Finally, a pullback-weight kN function gives the invariant section fe^{kN} , while a pullback-weight $-kN$ function gives the invariant section fe^{-kN} , under the inverse-character convention. Thus, after descent, degree one on S_+ gives $\mathcal{O}(1) = NL_+$, whereas degree one on S_- gives $\mathcal{O}(1) = -NL_-$. Clearing the finite stabilizer characters makes these honest line bundles and proves the last assertion. $\square$

Lemma 3.5. *Let E_c be the valuation divisor in (3.3). If $Q \subset X_c$ is a prime divisor whose generic point is contained in the image of the fixed locus at the wall, then*

$$\text{coeff}_Q(g_* E_c) > 0.$$

Proof. Any proper birational morphism between normal varieties is an isomorphism over the generic point of every prime divisor on the target. Localize the critical quotient at the generic point of Q ; its local ring R is a DVR, and both adjacent modifications are isomorphisms there. Let Z be a fixed component whose image $\bar{Z} \subset X_c$ contains the generic point of Q . By [HZ26, Proposition 6.9(ii)–(iii)], the attracting and repelling source strata of Z have dimension less than n , and their adjacent quotient strata both dominate $\bar{Z}$. Their orbits are one-dimensional because the adjacent stabilizers are finite, so those quotient strata have dimension at most $n - 2$. Hence

$$\dim \bar{Z} \leq n - 2 = \dim Q,$$

while the hypothesis that the generic point of Q lies in $\bar{Z}$ implies $Q \subset \bar{Z}$ and hence the reverse dimension inequality. Therefore $\bar{Z} = Q$. In particular both the positive- and negative-weight semi-invariant algebras below are nonzero.

Use the critical algebraization fixed in Lemma 3.2. Choose first an affine neighborhood of the generic point of Q in X_c and then localize its coordinate ring at Q , continuing to denote the resulting DVR by R . The saturated inverse image is affine, say $\text{Spec } A$, and the good-quotient identity gives $R = A_0$. Thus the local wall is described by the normal graded domain

$$A = \bigoplus_{j \in \mathbb{Z}} A_j, \quad R = A_0,$$

where the grading is the geometric-pullback grading fixed above,

$$A_j := \{a \in A \mid \rho_\lambda^* a = a \circ \rho_\lambda = \lambda^j a\}.$$

Lemma 3.4, followed by this localization, gives

$$X_- \times_{X_c} \text{Spec } R = \text{Proj}_R \bigoplus_{k \geq 0} A_{-kN}, \quad X_+ \times_{X_c} \text{Spec } R = \text{Proj}_R \bigoplus_{k \geq 0} A_{kN}.$$

Since the two half-algebras are finitely generated, after one sufficiently divisible common Veronese choose $N > 0$ so that

$$S_\pm = \bigoplus_{k \geq 0} A_{\pm kN}$$

are generated in degree one. Enlarge N simultaneously to clear the finite stabilizer characters. The same lemma identifies the tautological $\mathcal{O}(1)$ with $-NL_-$ on the lower relative Proj and with NL_+ on the upper relative Proj.

Choose $0 \neq h_- \in A_{-N}$ and set $I_+ = h_- A_N \subset A_0 = R$. Because S_+ is generated in degree one, $h_-^k A_{kN} = (h_- A_N)^k = I_+^k$. Hence multiplication by h_-^k identifies the k -th graded piece of S_+ with the k -th piece of the Rees algebra, and

$$\text{Proj}_R S_+ \simeq \text{Bl}_{I_+}(\text{Spec } R).$$

Since I_+ is a nonzero ideal of the DVR R , it is principal. Write $I_+ = (s_+)$. Since $s_+ = h_- f_+$ for some $f_+ \in A_N$, cancellation in the domain A gives $A_N = f_+ R$. Repeating the argument with the signs reversed gives $A_{-N} = f_- R$. Let $\mathfrak{m}_R$ be the maximal ideal of the DVR R ; it is the localized height-one prime corresponding to Q . Since Z is fixed, every semi-invariant of nonzero weight restricts to zero on Z : for $z \in Z$, the identity $f_\pm(z) = \lambda^{\pm N} f_\pm(z)$ for every $\lambda \in \mathbb{C}^*$ forces $f_\pm(z) = 0$. Hence the invariant function $\gamma_Q = f_+ f_- \in A_0 = R$ vanishes on Z . Since $\overline{\varpi_c(Z)} = Q$, it vanishes along Q , and therefore $\gamma_Q \in \mathfrak{m}_R$. Finally A is a domain, so $\gamma_Q \neq 0$; as R is a DVR, this gives $\text{ord}_Q(\gamma_Q) > 0$.

A pullback-weight- N function is a section of NL , so f_+ is a frame of NL_+ on the positive chamber. The function f_- has pullback weight $-N$, so f_-^{-1} is a frame of NL_- on the negative chamber. Choose the common rational frame $\ell^{(N)}$ in (3.3) to be f_+ . Its upper extension is the frame f_+ , whereas its lower extension is

$$f_+ = \gamma_Q f_-^{-1}.$$

The upper order is zero and the lower order is $\text{ord}_Q(\gamma_Q)$. Thus

$$N \text{coeff}_Q(g_* E_c) = \text{ord}_Q(\gamma_Q) > 0,$$

with the character convention fixed above. □

Lemma 3.6 (Oriented wall effectivity). *For the map g in diagram (1.1), orient the chambers so that $X_- = X_{c-\varepsilon}$ and $X_+ = X_{c+\varepsilon}$. Then the $\mathbb{Q}$ -Cartier divisor E_c defined in (3.3), namely $p_-^* L_- - p_+^* L_+$, is effective.*

Proof. Choose N and $A_{c,N}$ as in Lemma 3.2, choose a nonzero rational section s of $A_{c,N}$, and put $B_c = N^{-1} \text{div}(s)$. With the compatible rational representatives fixed in Proposition 3.3,

$$K_{X_\pm} + D_\pm - cL_\pm = \pi_\pm^* B_c.$$

The reduced classes in the two adjacent chambers are

$$\begin{aligned} H_- &:= -(K_{X_-} + D_-) + (c - \varepsilon)L_- = -\pi_-^* B_c - \varepsilon L_-, \\ H_+ &:= -(K_{X_+} + D_+) + (c + \varepsilon)L_+ = -\pi_+^* B_c + \varepsilon L_+. \end{aligned}$$

They are ample by Proposition 2.4, hence relatively ample for the respective modifications. Twisting by a pullback from the base does not change relative ampleness. The two displayed identities therefore show that $-L_-$ is π_- -ample and L_+ is π_+ -ample. Consequently

$$-E_c = -p_-^* L_- + p_+^* L_+$$

is g -nef. We next check the non-exceptional coefficients. Let Q be a prime divisor on X_c . Away from the image of the critical locus, both modifications identify a neighborhood of the generic point of Q with the common quotient open subset, and the coefficient of $g_* E_c$ along Q is zero. Suppose that Q is the image of a fixed stratum. Since Q has codimension one on the critical quotient, both modifications are isomorphisms over its generic point. The direct graded-algebra calculation in Lemma 3.5 gives

$$\text{coeff}_Q(g_* E_c) > 0.$$

Thus $g_* E_c$ is effective. We have shown that $-E_c$ is g -nef and $g_* E_c \geq 0$. Since W and X_c are normal and g is proper birational, the negativity lemma [KM98, Lemma 3.39] gives $E_c \geq 0$. $\square$

4. UNIFORM BOUNDS AND COMPACTNESS

The graph identity and $E_c \geq 0$ transport fixed-index log-Calabi–Yau boundaries across singular positive walls. Tracking the same principal divisor on every reduction then places the coefficients of $B_r = D_r + \Delta_r$ in one fixed finite rational set; this finite set is the coefficient input required by Hacon–Xu [HX15]. Then we prove a uniform lower bound on the normalized volumes, which implies bounded stabilizers and Cartier indices. Finally we use the relative length theorem [BCHM10] to obtain the bounded positive critical values, and a compact fixed locus.

4.1. Fixed-index boundaries across positive walls.

Lemma 4.1 (Fixed-index boundaries in every positive chamber). *There exist an integer $m > 0$ and a number $\delta > 0$, independent of the positive regular value r , and an effective big $\mathbb{Q}$ -divisor Δ_r on X_r such that*

$$(X_r, D_r + \Delta_r) \text{ is } \delta\text{-lc},$$

$$m(K_{X_r} + D_r + \Delta_r) \sim 0.$$

Moreover, the coefficients of $D_r + \Delta_r$ belong to one finite set independent of r .

Proof. Fix a nonzero rational top form κ on the common function field and, on every model, use the canonical divisor

$$K_{X_r} := \text{div}_{X_r}(\kappa).$$

Start with the complement on the first positive reduction from Proposition 2.7. Choose $m > 0$ sufficiently divisible and $\varphi \in K^\times$ so that

$$m(K_{X_-} + D_- + \Delta_-) = \text{div}_{X_-}(\varphi).$$

The same rational function φ is used on every later model.

Let $c > 0$ be such a wall and suppose that the complement has already been constructed on the chamber below it. On the chamber above it, define

$$\Delta_+ := \frac{1}{m} \text{div}_{X_+}(\varphi) - (K_{X_+} + D_+).$$

This divisor is $\mathbb{Q}$ -Cartier because $K_{X_+} + D_+$ is $\mathbb{Q}$ -Cartier by Proposition 2.4. On the normalized graph, put $E_c = p_-^* L_- - p_+^* L_+$. The wall identity (3.2) gives

$$p_+^*(K_{X_+} + D_+) = p_-^*(K_{X_-} + D_-) - cE_c.$$

The pullbacks of the principal divisor of φ agree. Therefore

$$p_+^* \Delta_+ = p_-^* \Delta_- + cE_c. \quad (4.1)$$

Pushing (4.1) forward by p_+ proves $\Delta_+ \geq 0$. Moreover $p_-^* \Delta_-$ is big and adding $cE_c \geq 0$ preserves bigness, so $p_+^* \Delta_+$ is big. Bigness is invariant under birational pullback, and therefore Δ_+ is big.

Since the same rational function φ is used on both models,

$$p_+^*(K_{X_+} + D_+ + \Delta_+) = \frac{1}{m} \operatorname{div}_W(\varphi) = p_-^*(K_{X_-} + D_- + \Delta_-). \quad (4.2)$$

Thus the two pairs are crepant birational. Pulling this identity to a model over W extracting any prescribed divisorial valuation shows that all their log discrepancies agree. Hence the initial value of δ works in every positive chamber. By construction,

$$m(K_{X_+} + D_+ + \Delta_+) = \operatorname{div}_{X_+}(\varphi),$$

so the complement index remains equal to m .

Finally set $B_r = D_r + \Delta_r$. The fixed canonical divisors $K_{X_r} = \operatorname{div}_{X_r}(\kappa)$ have integral coefficients. The equality $m(K_{X_r} + B_r) = \operatorname{div}_{X_r}(\varphi)$ then shows that every coefficient of B_r lies in $m^{-1}\mathbb{Z}$. Effectivity and δ -log canonicity give

$$0 \leq \operatorname{coeff}(B_r) \leq 1 - \delta.$$

Thus all coefficients lie in the finite set

$$m^{-1}\mathbb{Z} \cap [0, 1 - \delta],$$

independently of r . □

4.2. Bounded reductions and a uniform volume gap. Sun–Zhang apply Birkar’s coefficient-flexible [Bir21, Corollary 1.4] to bound the underlying reductions. Here the fixed finite rational coefficient set allows Hacon–Xu’s Theorem 1.3 to bound the marked pairs themselves.

Lemma 4.2 (Uniform reduction-volume gap). *There is a number $\nu > 0$ such that*

$$\widehat{\operatorname{vol}}(x, X_r, D_r) \geq \nu$$

for every positive regular value r and every closed point $x \in X_r$.

Proof. Put $B_r := D_r + \Delta_r$. By Lemma 4.1, the (X_r, B_r) are projective klt pairs of fixed dimension, B_r is big, $K_{X_r} + B_r \sim_{\mathbb{Q}} 0$, and the coefficients of B_r belong to one fixed finite set. Hence [HX15, Theorem 1.3] shows that these pairs form a bounded family.

Taking $\epsilon = \delta/2$, generic flatness, [HX15, Proposition 2.4], and Noetherian induction place them in finitely many $\mathbb{Q}$ -Gorenstein flat families of klt pairs. Base-changing each family by its total space and using the diagonal section gives pointed families containing every $x \in (X_r, B_r)$. By [BL21, Proposition 8 and Theorem 28], normalized volume is Zariski lower semicontinuous on each parameter space which gives a uniform constant $\nu > 0$ such that

$$\widehat{\operatorname{vol}}(x, X_r, B_r) \geq \nu.$$

Finally, for every valuation v centered at x , $A_{X_r, D_r}(v) = A_{X_r, B_r}(v) + v(\Delta_r) \geq A_{X_r, B_r}(v)$. Taking infima in the definition of normalized volume proves

$$\widehat{\operatorname{vol}}(x, X_r, D_r) \geq \widehat{\operatorname{vol}}(x, X_r, B_r) \geq \nu.$$

□

4.3. Stabilizers and a uniform Cartier index. Let $\nu > 0$ be the constant in Lemma 4.2. We bound the stabilizer order $|H|$ by comparing this uniform lower bound for $\widehat{\text{vol}}(x, X_r, D_r)$ with the universal upper bound on a transverse slice via the finite log-crepant formula.

Let $p \in X^s(\eta, r)$ be a stable point, let $x \in X_r$ be its image, and let

$$H := \text{Stab}_{\mathbb{C}^*}(p).$$

After restricting over an affine neighborhood of x , Luna’s algebraic étale slice theorem [Lun73] gives a transverse slice (S, s) and a strongly étale chart

$$\mathbb{C}^* \times_H S \longrightarrow X^s(\eta, r).$$

Since $X^s(\eta, r)$ is an open subspace of the klt space X , preservation of klt singularities under étale pullback along the two étale maps $\mathbb{C}^* \times S \rightarrow \mathbb{C}^* \times_H S \rightarrow X^s(\eta, r)$ shows that $\mathbb{C}^* \times S$ is klt; smooth-factor invariance then shows that S is klt. On the quotient, the local chain is

$$(s \in (S, 0)) \xrightarrow{\pi} (\bar{s} \in (S/H, D_H)) \xrightarrow[\text{étale}]{\tau} (x \in (X_r, D_r)).$$

Let $\pi: S \rightarrow S/H$ be the quotient map, and characterize D_H by

$$K_S = \pi^*(K_{S/H} + D_H).$$

The strongly étale Cartesian square and the ramification formula give

$$D_H = \tau^* D_r.$$

Thus D_H is the effective branch boundary of the finite quotient π .

Lemma 4.3 (Slice quotient formula). *The slice and the reduction have dimension $n - 1$, and*

$$|H| \widehat{\text{vol}}(x, X_r, D_r) = \widehat{\text{vol}}(s, S) \leq (n - 1)^{n-1}. \quad (4.3)$$

Proof. Since π is finite and τ is étale, both S and X_r have dimension $n - 1$. We first show that H acts faithfully on the slice. If $h \in H$ acts trivially on the slice germ, shrink S so that it fixes S pointwise. Since h fixes p and $\mathbb{C}^*$ is abelian, it also fixes the orbit $\mathbb{C}^* \cdot p$ pointwise. The Luna product chart then shows that h fixes a nonempty open subset of the irreducible space X . Hence h acts trivially on X , and effectivity of the $\mathbb{C}^*$ -action gives $h = 1$.

Thus a general point of S has trivial stabilizer, so the quotient $\pi: S \rightarrow S/H$ has Galois degree $|H|$. Since s is fixed by H , the fiber over $\bar{s}$ is supported at s , and π induces a finite Galois morphism of pointed germs of degree $|H|$. The source is klt by the slice argument above, while the target is klt because τ is étale and (X_r, D_r) is klt. The identity $K_S = \pi^*(K_{S/H} + D_H)$ makes this morphism log crepant; hence [XZ21, Theorem 1.3] gives

$$\widehat{\text{vol}}(s, S) = |H| \widehat{\text{vol}}(\bar{s}, S/H, D_H).$$

Finally, τ and $D_H = \tau^* D_r$ identify the completed pointed log germs at $\bar{s}$ and x , so

$$\widehat{\text{vol}}(\bar{s}, S/H, D_H) = \widehat{\text{vol}}(x, X_r, D_r)$$

by [BL21, Theorem 5] and [dFEM11, Proposition 2.11]. The universal bound [LX19, Theorem A.4] gives $\widehat{\text{vol}}(s, S) \leq (n - 1)^{n-1}$, and the result follows. $\square$

In the smooth setting of Sun–Zhang, the weight–gcd argument bounds the pseudoreflexion contribution and isolates a residual quotient singularity of the underlying coarse variety, to which ordinary normalized volume applies [SZ26, Section 3, especially Lemma 3.6]. Here a Luna slice is only klt and can ramify divisorially, so we retain D_r and use the finite log-crepant formula directly; it controls the full stabilizer H in one step.

Lemma 4.4. *There is an integer $M > 0$ such that ML_r is a Cartier line bundle for every positive regular value r .*

Proof. Let $\nu > 0$ be the constant in Lemma 4.2, let p be any stable point over a positive regular reduction, let $x \in X_r$ be its image, and put $H = \text{Stab}_{\mathbb{C}^*}(p)$. By (4.3),

$$|H| \nu \leq |H| \widehat{\text{vol}}(x, X_r, D_r) \leq (n-1)^{n-1}.$$

Thus $|H| \leq (n-1)^{n-1}/\nu$, and every stabilizer order is at most

$$B := \max \left\{ 1, \left\lfloor \frac{(n-1)^{n-1}}{\nu} \right\rfloor \right\}.$$

Set $M = \text{lcm}\{1, \dots, B\}$.

Every stabilizer acts trivially on the fiber of $\widetilde{L}_r^{\otimes M}$. By the descent theorem [KKV89, §4, Proposition 4.2, pp. 83–84], this power descends to a line bundle on X_r . By the definition and uniqueness of L_r , its descent is ML_r . $\square$

4.4. Bounded critical values and compactness.

Lemma 4.5. *Let c be an isolated critical value with both adjacent levels nonempty. If both adjacent modifications are isomorphisms, then E_c has a nonzero positive coefficient.*

Proof. Fix a general closed point q in the image of a fixed component Z and localize the critical quotient at q . In the notation of the proof of Lemma 3.5, we have a normal graded domain

$$A = \bigoplus_{j \in \mathbb{Z}} A_j, \quad R = \mathcal{O}_{X_c, q} = A_0, \quad S_{\pm} = \bigoplus_{k \geq 0} A_{\pm kN},$$

where both signs occur by [HZ26, Proposition 6.9] and a common Veronese makes $S_{\pm}$ generated in degree one. The two relative Proj morphisms are isomorphisms by hypothesis.

Choose $0 \neq h_- \in A_{-N}$. The Rees-algebra calculation in the proof of Lemma 3.5 identifies $\text{Proj}_R S_+$ with the blowup of $h_- A_N$. Hence this ideal is invertible, and therefore principal because R is local. Cancellation in A gives $A_N = f_+ R$; reversing the signs gives $A_{-N} = f_- R$.

As in that proof, $\gamma_Z := f_+ f_-$ is nonzero and lies in the maximal ideal of R , since nonzero-weight semi-invariants vanish on Z . The same frame comparison gives

$$NE_c = \text{div}_R(\gamma_Z).$$

A nonzero nonunit in the normal local domain R has positive order along some prime divisor, proving the claim. $\square$

Let M be the integer in Lemma 4.4.

Proposition 4.6 (Uniform bound for positive critical values). *Every positive critical value satisfies*

$$c \leq 2(n-1)M.$$

Therefore, the zero set of η is compact.

Proof. Fix a positive wall c and retain the notation

$$\pi_{\pm}: X_{\pm} \longrightarrow X_c, \quad E_c = p_-^* L_- - p_+^* L_+.$$

Since $ML_{\pm}$ are Cartier, ME_c is an integral Cartier divisor. Wall effectivity makes it effective, so every nonzero coefficient a of E_c satisfies $a \geq 1/M$.

First suppose that π_- is not an isomorphism. Over X_c one has

$$K_{X_-} + D_- \equiv_{\pi_-} cL_-,$$

and $-L_-$ is π_- -ample. Hence $-(K_{X_-} + D_-)$ is π_- -ample. The relative length theorem [BCHM10, Theorem 3.8.1], applied to the klt pair (X_-, D_-) over X_c , gives a contracted rational curve C with

$$0 < -(K_{X_-} + D_-) \cdot C \leq 2(n-1).$$

Since ML_- is Cartier and $-L_-$ is relatively ample,

$$-ML_- \cdot C \in \mathbb{Z}_{>0}, \quad -L_- \cdot C \geq \frac{1}{M}.$$

Intersecting the critical descent identity with C gives

$$c = \frac{-(K_{X_-} + D_-) \cdot C}{-L_- \cdot C} \leq 2(n-1)M.$$

It remains to assume that π_- is an isomorphism. Identify $X_- = X_c$ via π_- . Then we can take normalized graph W to be X_+ and therefore

$$E_c = \pi_+^* L_- - L_+.$$

The proof of Lemma 3.6 shows that L_+ is π_+ -ample, so $-E_c = L_+ - \pi_+^* L_-$ is also π_+ -ample. If π_+ is not an isomorphism, then $E_c \neq 0$. Choosing a prime component F of E_c , we have

$$a := \text{coeff}_F(E_c) \geq \frac{1}{M}.$$

The wall identity becomes

$$K_{X_+} + D_+ = \pi_+^*(K_{X_-} + D_-) - cE_c.$$

Since D_+ is a boundary, $d_F := \text{coeff}_F(D_+) \geq 0$. Since (X_-, D_-) is klt, the log discrepancy $A_F := A(F; X_-, D_-)$ is positive. Using compatible canonical divisors, the definition of log discrepancy and the wall identity give

$$\begin{aligned} A_F &= 1 + \text{coeff}_F(K_{X_+} - \pi_+^*(K_{X_-} + D_-)) \\ &= 1 - d_F - ca. \end{aligned}$$

Since $a \geq 1/M$, $A_F > 0$, and $d_F \geq 0$, it follows that $c < M$.

Finally suppose that both modifications are isomorphisms. Lemma 4.5 gives $E_c \neq 0$. On the common normal variety choose the same canonical divisor on both sides; the canonical terms then cancel and the wall identity becomes

$$D_- - D_+ = cE_c.$$

For a component F of E_c , again $a = \text{coeff}_F(E_c)$, while both boundary coefficients belong to $[0, 1)$. Hence

$$ca = \text{coeff}_F(D_-) - \text{coeff}_F(D_+) < 1,$$

and $c < M$. These cases exhaust the wall.

Because $\eta = J\nabla u$, the set $Z(\eta)$ is the critical locus of u . The bound just proved and the lower bound for u place $u(Z(\eta))$ in a compact interval. Since u is proper, $Z(\eta)$ is compact. $\square$

5. CONSTRUCTION OF THE FANO FIBRATION

Following the smooth blueprint of Sun–Zhang, we construct the punctured-cone map and, in the singular noncompact setting, extend it across the compact complement, prove properness, globalize a fixed anticanonical embedding, algebraize it by GAGA, and verify Reeb positivity.

Lemma 5.1 (Stabilized quotient and its punctured cone). *There exist a regular value r_* , a $\mathbb{C}^*$ -invariant Zariski-open set $U \subset X$, a normal projective klt pair (M, D) , a semi-ample $\mathbb{Q}$ -line bundle L_0 on M , and a proper surjective algebraic equivariant map*

$$\pi_U: U \longrightarrow Y' \setminus \{o\},$$

where Y' is a normal affine cone with vertex o . The set U and the quotient (M, D, L_0) are the stable set and polarized quotient at every regular value $r \geq r_*$.

Proof. By Proposition 4.6, the critical values are bounded above; choose a regular r_* larger than all of them. Then the stable set and its quotient are independent of $r \geq r_*$. We denote them by

$$U = X^s(\eta, r), \quad (M, D, L_0).$$

By [HZ26, Theorem 6.6 and the final paragraph of its proof], U is a $\mathbb{C}^*$ -invariant Zariski-open quasi-projective variety, $U \rightarrow M$ is an algebraic good quotient, and M is normal and projective. The pair (M, D) is klt by Proposition 2.4. Formula (2.4) becomes

$$-(K_M + D) + rL_0 \in \text{Amp}(M), \quad r \geq r_*.$$

Dividing by r and letting $r \rightarrow \infty$ shows that L_0 is nef, while $aL_0 - (K_M + D)$ is ample for $a \gg 0$. Applied to a Cartier multiple of L_0 , the base-point-free theorem [KM98, Theorem 3.3] makes L_0 semi-ample.

Choose $k > 0$, divisible by the uniform index of Lemma 4.4, such that $L := kL_0$ is a base-point-free Cartier line bundle. Let

$$p: M \longrightarrow M', \quad p^*L' = L,$$

be its semi-ample morphism with connected fibers. After replacing k by a multiple, we may assume that L' is very ample and projectively normal. The $\mathbb{T}$ -linearization of the character line descends to L , so p is equivariant. Since

$$H^0(M, L^j) = H^0(M', jL'),$$

a finite set of homogeneous eigengenerators gives algebraic $\mathbb{T}^{\mathbb{C}}$ -actions on M' and its section ring. Set

$$Y' := \text{Spec} \bigoplus_{j \geq 0} H^0(M', jL')$$

and denote its vertex by o .

The quotient $U \rightarrow M$ is the punctured orbibundle of the character line $L_0^\vee$. On a slice with stabilizer H , the k -th fiber-power map is

$$\mathbb{C}^* \times_H S \longrightarrow (S/H) \times \mathbb{C}^*, \quad [t, s] \longmapsto (\bar{s}, t^k).$$

Because L is the descent of the k -th power of the character line, H acts trivially on that power, so $h^k = 1$ for every $h \in H$. The map is therefore well defined and finite because both $S \rightarrow S/H$ and $t \mapsto t^k$ are finite. These slice maps glue to a finite surjective map from U to the punctured bundle of $L^\vee = p^*(L')^\vee$. This bundle maps properly to the punctured bundle of $(L')^\vee$. Since L' is very ample and projectively normal, the latter is $Y' \setminus \{o\}$. The composite is the required map π_U . $\square$

Put $A := X \setminus U$. The affine property of Y' and the equivariant property of π_U and the properness of the function u implies that π_U extends uniquely to a proper surjective $\mathbb{T}$ -equivariant holomorphic map

$$\pi': X \longrightarrow Y'.$$

Moreover, A is compact analytic and $\pi'^{-1}(o) = A$.

Lemma 5.2 (Equivariant anticanonical algebraization). *There are $m > 0$ and $N > 0$ such that $-mK_X$ is Cartier, and finitely many $\mathbb{T}$ -eigensections of $\mathcal{O}_X(-mK_X)$ define a $\mathbb{T}$ -equivariant closed analytic immersion*

$$(\pi', F): X \hookrightarrow Y' \times \mathbb{P}^N.$$

Moreover,

$$\mathcal{O}_X(-mK_X) \simeq F^* \mathcal{O}_{\mathbb{P}^N}(1),$$

X is the analytification of a normal quasi-projective variety, π' is projective algebraic, and $-K_X$ is π' -ample.

Proof. Choose relatively compact $\mathbb{T}$ -invariant Stein neighborhoods $B \subseteq B'$ of o , stable under the contracting radial semigroup, and put $V' := \pi'^{-1}(B')$. Compactness of $K' := \pi'^{-1}(\overline{B'})$ gives $m_0 > 0$ such that $L := \mathcal{O}_X(-m_0 K_X)$ is a line bundle near K' . Its non-locally-free locus is radial-invariant, and every radial orbit meets K' ; hence L is a line bundle on all of X .

The soliton metric on $L|_{X^{\text{reg}}}$ has curvature $m_0 \omega$. Standard Richberg regularization makes L positive in Grauert's sense, so it is relatively ample for π' [Gra62, §3, Satz 2 and its proof]. For a sufficiently divisible m , the relative Kodaira–Grauert theorem [CDJ13, pp. 859–860] gives finitely many sections of $\mathcal{O}_X(-m K_X)$ on V' whose projectivization, together with π' , is a closed immersion over B .

Approximating these sections and their first jets on $\pi'^{-1}(\overline{B})$ by finite Fourier sums, and then retaining every weight component, gives $\mathbb{T}$ -eigensections $s_0, \dots, s_N$ with the same separation property. Indeed, the Fourier truncations are linear combinations of the component coordinates. Homogeneity extends the s_i uniquely along radial orbits, and propagates base-point-freeness and point and jet separation from $\pi'^{-1}(B)$ to X . Thus

$$F = [s_0 : \dots : s_N], \quad (\pi', F): X \hookrightarrow Y' \times \mathbb{P}^N$$

is an injective local closed immersion. It is proper because π' is proper, hence is a closed analytic immersion, and

$$\mathcal{O}_X(-m K_X) \simeq F^* \mathcal{O}_{\mathbb{P}^N}(1).$$

Let Z be its analytic image, let $q: Y' \times \mathbb{P}^N \rightarrow Y'$ be the projection, and let $\mathcal{I}_Z$ be the ideal of Z . On a small invariant Stein neighborhood V of o , relative Cartan–Serre generation and Grauert coherence make $\mathcal{E}_d := q_* \mathcal{I}_Z(d)$ coherent and its evaluation surjective for $d \gg 0$ [GPR94, Chapter III, Section 4.1 and Chapter V, Section 4.8]. Fourier projection and equivariant Nakayama give homogeneous generators

$$G_\nu(y, w) = \sum_{|\alpha|=d} f_{\nu, \alpha}(y) w^\alpha.$$

Each coefficient is a fixed-weight holomorphic germ at o . Such a germ is regular: after embedding Y' by homogeneous generators of strictly positive weights, the corresponding weight component of an ambient Taylor series contains only finitely many monomials. Hence the G_ν extend to homogeneous algebraic equations on $Y' \times \mathbb{P}^N$.

Let $\mathcal{J}$ be the algebraic ideal they generate and put $W := V(\mathcal{J})$. By construction, $\mathcal{J}^{\text{an}} = \mathcal{I}_Z$ over $q^{-1}(V)$. Both ideals are radial-invariant, and every radial orbit meets V , so this equality holds globally. Thus $W^{\text{an}} = Z$ scheme-theoretically. Since $Z \simeq X$ is normal, analytification detects normality [SGA1, Exposé XII, Proposition 2.1(vi)]; hence W is normal.

We identify X with the normal quasi-projective variety $W \subset Y' \times \mathbb{P}^N$. Then (π', F) is algebraic and π' is projective. Moreover, the algebraic stabilizer of W in $\mathbb{T}^{\mathbb{C}}$ contains the Zariski-dense subgroup $\mathbb{T}$, so the algebraization is $\mathbb{T}^{\mathbb{C}}$ -equivariant.

Finally, reflexive canonical powers commute with analytification on normal varieties, and therefore

$$(\mathcal{O}_W(-m K_W))^{\text{an}} \simeq \mathcal{O}_X(-m K_X).$$

Faithful flatness makes $\mathcal{O}_W(-m K_W)$ a line bundle [SGA1, Exposé XII, Theorem 1.1]. On every projective fiber of π' , proper GAGA [SGA1, Exposé XII, Theorem 4.4] identifies it with $F^* \mathcal{O}_{\mathbb{P}^N}(1)$, hence it is ample. The fiberwise criterion [EGAIII₁, Théorème 4.7.1] makes $\mathcal{O}_W(-m K_W)$ π' -ample. Thus $-K_W$ is π' -ample. $\square$

Let

$$X \xrightarrow{\pi} Y \xrightarrow{\nu} Y'$$

be the $\mathbb{T}$ -equivariant algebraic Stein factorization of π' .

Lemma 5.3 (Positivity of the soliton weights). *Every nonconstant $\mathbb{T}$ -eigenfunction $g \in \mathcal{O}(Y)_\alpha$ satisfies $\langle \alpha, \xi \rangle > 0$.*

Proof. Since $\mathbb{T}$ is the closure of the ξ -flow and $\eta \in \text{Lie}(\mathbb{T})$, one has $Z(\xi) \subset Z(\eta)$; the latter is compact by Proposition 4.6. Choose $b > \max_{Z(\xi)} f$. If Φ_t is the complete flow of $-J\xi = \nabla f$. For $0 \neq g \in \mathcal{O}(Y)_\alpha$, set $\tilde{g} := \pi^*g$ and $\lambda := \langle \alpha, \xi \rangle$. Equivariance gives

$$\tilde{g}(\Phi_t x) = e^{\lambda t} \tilde{g}(x).$$

If $\lambda \leq 0$, then $|\tilde{g}|$ attains its global maximum on the compact set $\{f \leq b\}$, so the maximum-modulus principle makes $\tilde{g}$ constant. Since π is surjective, g is constant. Hence every nonconstant eigenfunction has $\lambda > 0$. $\square$

Proof of Theorem 1.1. Lemmas 5.1–5.2 give a normal quasi-projective algebraic structure on X , a projective morphism $\pi': X \rightarrow Y'$, and a π' -ample anticanonical class $-K_X$. For the Stein factorization above, Y is normal, ν is finite, and $\pi_*\mathcal{O}_X = \mathcal{O}_Y$. Since Y' is affine, the finiteness of ν implies that Y is affine [Har77, Chapter II, Exercise 3.4]. The graph of π is a closed immersion

$$X \longrightarrow X \times_{Y'} Y,$$

and the target is projective over Y ; hence π is projective. A Cartier multiple of $-K_X$ is π' -ample and therefore π -ample by [EGAII, Proposition 4.6.13(v), pp. 91–92]. Analytic and algebraic klt singularities agree over $\mathbb{C}$ [Kaw24, Proposition 1.10.3(3)]. Hence $\pi: X \rightarrow Y$ is a Fano fibration. The $\mathbb{T}$ -action descends through the canonical Stein factorization, so all maps are equivariant.

Put $R = \mathcal{O}(Y)$. Lemma 5.3 gives $R_0 = \mathbb{C}$ and $\langle \alpha, \xi \rangle > 0$ for every nonzero weight occurring in R . Consequently, $\bigoplus_{\alpha \neq 0} R_\alpha$ is the unique $\mathbb{T}$ -fixed maximal ideal of R , so Y has a unique $\mathbb{T}$ -fixed point and ξ lies in the Reeb cone. Therefore $(\pi: X \rightarrow Y, \xi)$ is a polarized Fano fibration. $\square$

REFERENCES

- [BCHM10] C. Birkar, P. Cascini, C. D. Hacon, and J. McKernan. Existence of minimal models for varieties of log general type. *J. Amer. Math. Soc.*, 23(2):405–468, 2010.
- [Bir21] C. Birkar. Singularities of linear systems and boundedness of Fano varieties. *Ann. of Math. (2)*, 193(2):347–405, 2021.
- [BL21] H. Blum and Y. Liu. The normalized volume of a singularity is lower semicontinuous. *J. Eur. Math. Soc.*, 23(4):1225–1256, 2021.
- [CMZ24] P.-Y. Chan, Z. Ma, and Y. Zhang. On noncollapsed $\mathbb{F}$ -limit metric solitons. Preprint, arXiv:2401.03387, 2024.
- [CDJ13] M. Coltoiu, K. Diederich, and C. Joița. On complex spaces with prescribed singularities. *Math. Res. Lett.*, 20(5):857–868, 2013.
- [dFEM11] T. de Fernex, L. Ein, and M. Mustața. Log canonical thresholds on varieties with bounded singularities. In *Classification of algebraic varieties*, EMS Ser. Congr. Rep., pages 221–257. European Mathematical Society, Zürich, 2011.
- [Dem] J.-P. Demailly. *Complex analytic and differential geometry*. Online book.
- [DS17] S. K. Donaldson and S. Sun. Gromov–Hausdorff limits of Kähler manifolds and algebraic geometry, II. *J. Differential Geom.*, 107(2):327–371, 2017.
- [EGAII] A. Grothendieck and J. Dieudonné. *Éléments de géométrie algébrique. II. Étude globale élémentaire de quelques classes de morphismes*. *Publ. Math. Inst. Hautes Études Sci.*, 8:5–222, 1961.
- [EGAIII₁] A. Grothendieck. *Éléments de géométrie algébrique. III. Étude cohomologique des faisceaux cohérents. I*. *Publ. Math. Inst. Hautes Études Sci.*, 11:5–167, 1961.
- [Gra62] H. Grauert. Über Modifikationen und exzeptionelle analytische Mengen. *Math. Ann.*, 146:331–368, 1962.
- [Gre10] D. Greb. Projectivity of analytic Hilbert and Kähler quotients. *Trans. Amer. Math. Soc.*, 362(6):3243–3271, 2010.
- [GPR94] H. Grauert, T. Peternell, and R. Remmert, editors. *Several complex variables VII: Sheaf-theoretical methods in complex analysis*. Encyclopaedia of Mathematical Sciences, 74. Springer-Verlag, Berlin, 1994.
- [Har77] R. Hartshorne. *Algebraic geometry*. Graduate Texts in Mathematics, 52. Springer-Verlag, New York, 1977.
- [HZ26] M. Hallgren and J. Zhang. Singular Kähler–Ricci shrinkers are complex analytic. Preprint, arXiv:2605.25213, 2026.
- [HX15] C. D. Hacon and C. Xu. Boundedness of log Calabi–Yau pairs of Fano type. *Math. Res. Lett.*, 22(6):1699–1716, 2015.

- [Kaw24] Y. Kawamata. *Algebraic varieties: Minimal models and finite generation*. Translated by C. Jiang. Cambridge Studies in Advanced Mathematics, 214. Cambridge University Press, Cambridge, 2024.
- [KKV89] F. Knop, H. Kraft, and T. Vust. The Picard group of a G -variety. In H. Kraft, P. Slodowy, and T. A. Springer, editors, *Algebraische Transformationsgruppen und Invariantentheorie*, DMV Seminar, vol. 13, pages 77–87. Birkhäuser, Basel, 1989. https://doi.org/10.1007/978-3-0348-7662-9_5.
- [KM98] J. Kollár and S. Mori. *Birational geometry of algebraic varieties*. Cambridge Tracts in Mathematics, 134. Cambridge University Press, 1998.
- [LX19] Y. Liu and C. Xu. K-stability of cubic threefolds. *Duke Math. J.*, 168(11):2029–2073, 2019.
- [Lun73] D. Luna. Slices étales. In *Sur les groupes algébriques*, pages 81–105. Bull. Soc. Math. France, Mémoire 33, 1973.
- [SZ26] S. Sun and J. Zhang. Kähler–Ricci shrinkers and Fano fibrations, I. Preprint, arXiv:2410.09661, 2026.
- [SGA1] M. Raynaud. Géométrie algébrique et géométrie analytique. In A. Grothendieck, *Revêtements étales et groupe fondamental (SGA 1)*, Exposé XII, Lecture Notes in Mathematics, vol. 224, pages 311–343. Springer-Verlag, Berlin–New York, 1971.
- [SP] The Stacks Project Authors. *Stacks Project*. <https://stacks.math.columbia.edu>, 2026.
- [Tha96] M. Thaddeus. Geometric invariant theory and flips. *J. Amer. Math. Soc.*, 9(3):691–723, 1996. <https://doi.org/10.1090/S0894-0347-96-00204-4>.
- [XZ21] C. Xu and Z. Zhuang. Uniqueness of the minimizer of the normalized volume function. *Camb. J. Math.*, 9(1):149–176, 2021.

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